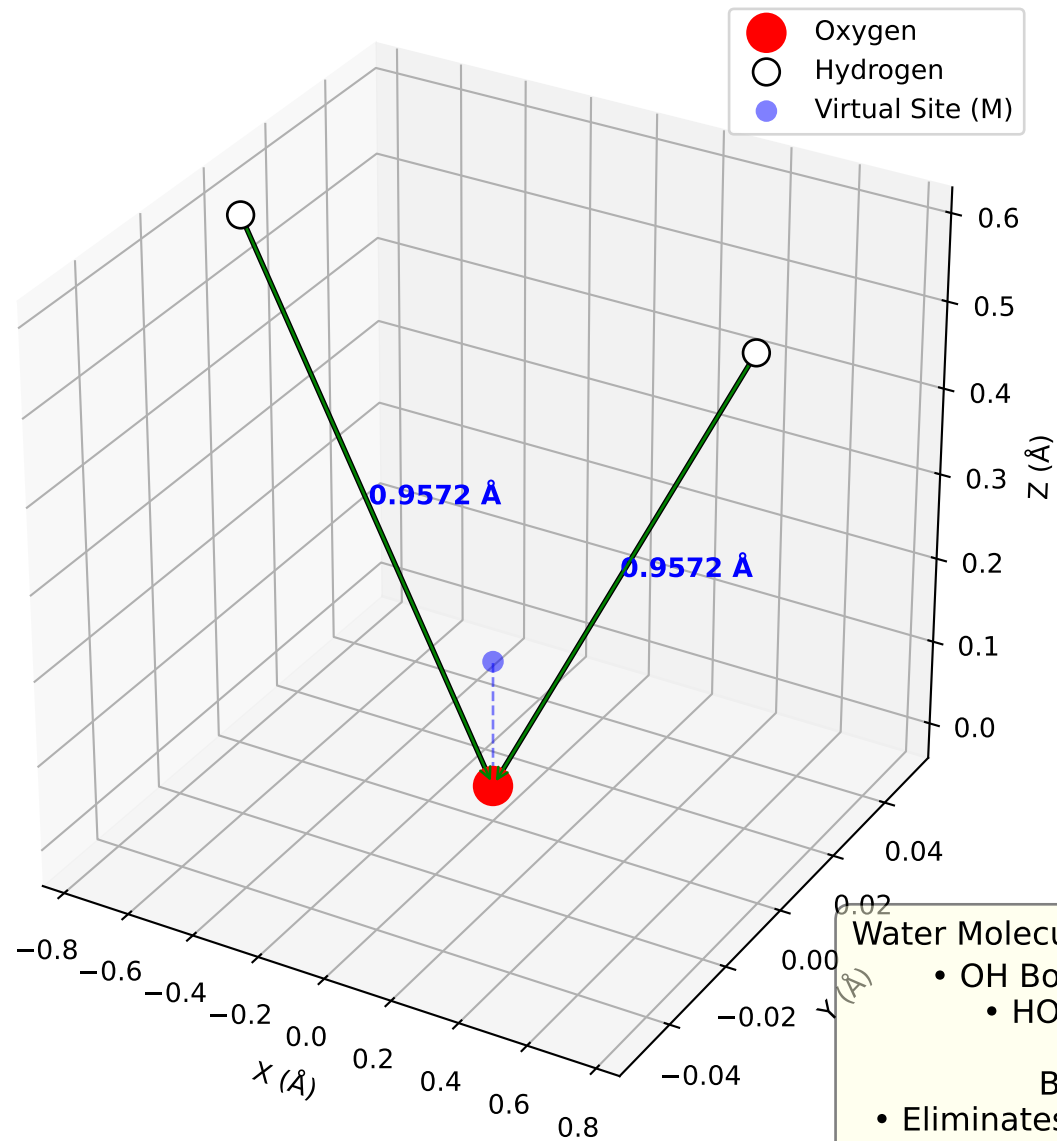
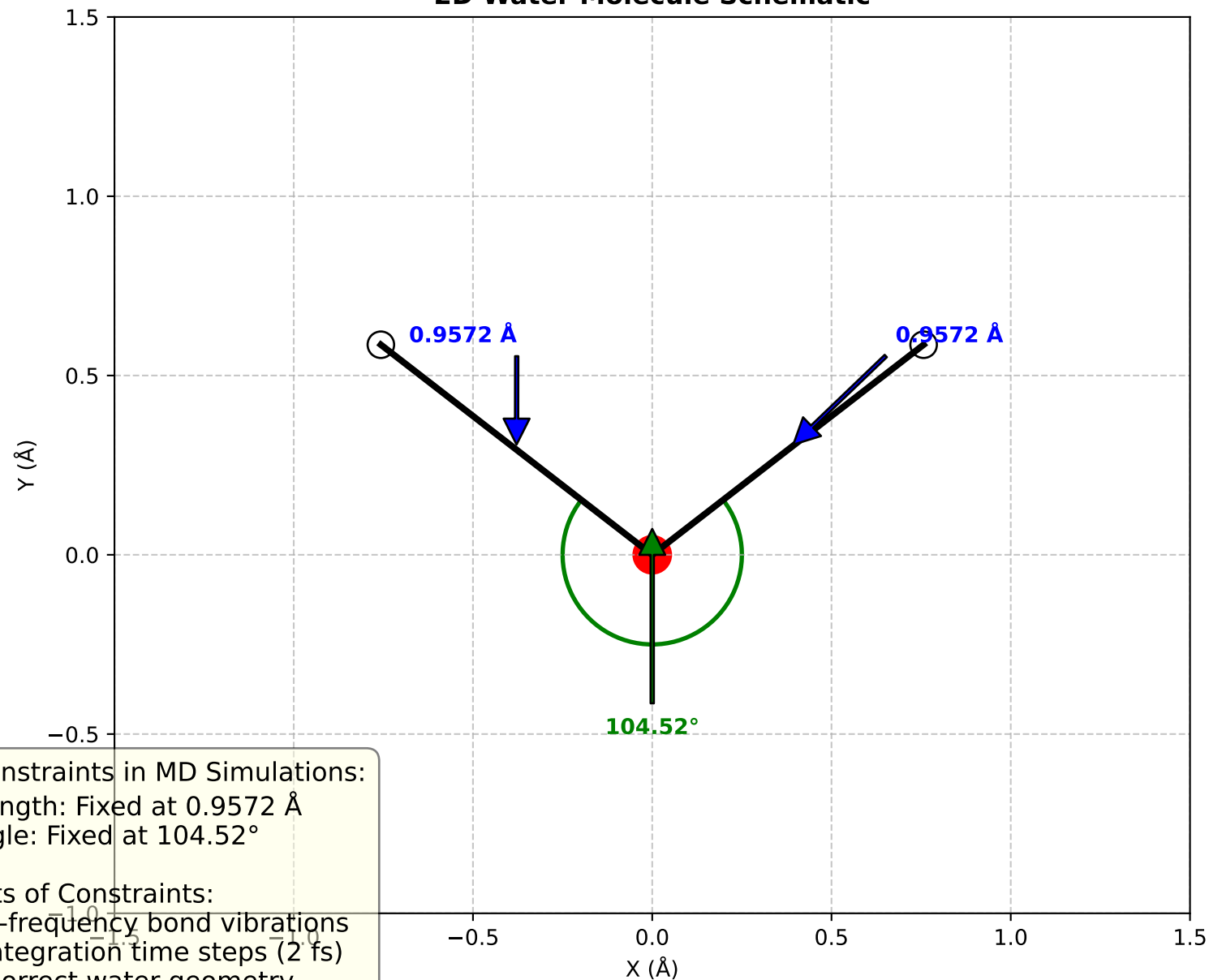


3D Water Molecule with Constraints



2D Water Molecule Schematic



Water Molecule Constraints in MD Simulations:

- OH Bond Length: Fixed at 0.9572 Å
- HOH Angle: Fixed at 104.52°

Benefits of Constraints:

- Eliminates high-frequency bond vibrations
- Allows larger integration time steps (2 fs)
 - Maintains correct water geometry
- Implemented using SHAKE/LINCS algorithms